Project 2: Virtual Pet

Advanced Embedded Systems | Team Plan for 5 Members | Due October 12, 2026

Guiding Principle

Each member owns one complete subsystem from start to finish: they write the code, commit it on their own feature branch, and explain it during the presentation. This matches what Dr. Shu said he will look for: every member making real commits on GitHub, every member speaking, and every member able to explain their part and the decisions behind it.

Role Summary

Member	Subsystem	Presents
1	Display and user interface	How the LCD works and how data is sent over SPI
2	Pet artwork and animation engine	Sprites, frames, memory budget
3	Pet logic and timing	State diagram, stats, responsiveness during animations
4	Sensors (ADC and I2C)	Raw to useful values, accelerometer addressing and reads
5	Outputs, hardware, and integration	Block diagram, pin plan, speaker, integration

Detailed Responsibilities

Member 1: Display and User Interface

- Run the instructor's LCD starter project and pass the smoke test.
- Build the graphics test screen (required by the assignment; keep it in the repo).
- Main pet screen, hunger and happiness meters, menu with visible selection, return to main screen.
- Redraw only the regions that change, to avoid flicker and reduce SPI traffic.

Member 2: Pet Artwork and Animation Engine

- Design the original pet and all frames: idle, feed, play, third action, death, runaway.
- Convert images to RGB565 arrays; check color order, orientation, and Flash usage (a 64 x 64 frame is about 8 KB).
- Write the animation engine code: frame tables and time-based frame switching without blocking. This gives this member solid technical commits, not only artwork.

Member 3: Pet Logic and Timing

- State machine: Idle, Menu, Feeding, Playing, Third Action, Dead, Ran Away.
- Hunger and happiness clamped to documented limits; decay over time; effect of each action.
- Death and runaway conditions with a clear restart procedure.
- Non-blocking main loop scheduled with HAL_GetTick() (no HAL_Delay() in the loop).

Member 4: Sensors

- Force sensor on the ADC: calibration, raw value (0 to 4095), conversion to a meaningful quantity. The video must show both the raw and the interpreted value.
- Accelerometer on I2C (borrow from Dr. Shu): read the ID register first to prove communication, then detect shaking by threshold.

Member 5: Outputs, Hardware, and Integration

- Solder the male headers and attach the display board correctly.
- Pin planning in the starter project's .ioc file, avoiding conflicts with the LCD, SPI, and joystick.
- Speaker with timer PWM (borrow from Dr. Shu): different sounds for happy, hungry, and dying.
- Optional creative output, such as an RGB LED showing the pet's mood.
- Repository management: main and develop branches, reviewing and merging pull requests, README.

Key to Working in Parallel

In the first meeting, agree on the interface between subsystems: which functions each part provides to the others. For example, Member 4 provides simple functions such as *"was the sensor squeezed?"* and *"was the board shaken?"*, and Member 3 only calls those. Until the sensors are ready, Member 3 can test with the joystick instead. This way nobody waits on anybody else.

All custom code goes in separate .c/.h files or inside USER CODE BEGIN / END blocks, so regenerating from CubeMX never erases anyone's work.

GitHub Workflow (as Dr. Shu requested)

- **main** branch: only the complete, working version.
- **develop** branch: where everyone's work is combined.
- Each member creates a feature branch for each feature, commits there, and merges into **develop** through a pull request after testing.
- Do not have one person commit everything on one branch; Dr. Shu will check branch activity.
- Keep the repository public so branch activity is visible. Commit early and often with descriptive messages.
- A README is required: pet, wiring, controls, structure, build steps, members, contributions, AI use.

Timeline

Deadline	Milestone
Oct 2	Headers soldered, starter project running, repo and branches created, interfaces agreed
Oct 7	Each subsystem complete and tested on its own branch
Oct 10	Everything merged into develop and debugged
Oct 10 to 11	Record video, build slides, rehearse presentation
Oct 12	Submit slides (one submission for the group)

Shared Tasks

- **Slides:** each member builds the slides for their own subsystem. Include the repo link and the embedded demo video.
- **Presentation:** at least 5 minutes, and every member must speak about what they built (roughly one minute each). Be ready for a live demo too; the video is the backup.
- **Video:** one person records, but each member checks that their items from the 12-item video list in the assignment are shown clearly.
- **AI use:** each member collects one example of how AI helped and one case where an AI suggestion had to be corrected or rejected.
- **Parts:** borrow the accelerometer and speaker from Dr. Shu soon (limited supply). Double-row male headers are easier to solder than the single-row ones he provides.